UPLOAD SLOT 02 OF 12

Neighborhood Park Preliminary Design Masterplan

The plan at scale. Every room struck off the crescent's centre; every area the measured area of the drawn polygon.

CONTENTS

Al Safa 2 Park — Falaj Al Safa

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Generated 2026-08-01 · reproducible: python tools/build_submission_pdfs.py

Master Plan Development

Al Safa 2 Park — Concept A "Shaded Spine" Implemented as Scaled Geometry

AI-GENERATED DRAFT — FOR REVIEW. This master plan implements Concept A ("Shaded Spine", Phase 4.7) as an actual scaled geometric layout — every zone is sized in real square meters, and the total sums exactly to the Brief's confirmed 15,000 sqm site area. Site proportions (150m x 100m) are an assumed elongated rectangle consistent with the aerial massing seen in the Phase 1 master-plan graphic; the exact boundary polygon awaits DWG conversion (logged gap).

5.1 Spatial Framework & 5.2 Functional Zoning

Figure 1 — Preliminary Master Plan, Concept A "Shaded Spine" (to scale, 15,000 sqm).

Zoning Area Schedule (computed, sums to 15,000 sqm)

Zone	Category	Area	% of Site
Main Entrance Plaza	Arrival	240 sqm	1.6%
Shaded Spine (Central Walkway)	Circulation	1,260 sqm	8.4%
Secondary Entrance (E)	Arrival	240 sqm	1.6%
Children's Play Zone	Active	1,088 sqm	7.3%
Family Picnic & Shaded Seating	Passive	884 sqm	5.9%
Community Plaza & Event Lawn	Social	1,224 sqm	8.2%
Outdoor Fitness & Wellness	Active	816 sqm	5.4%

Zone	Category	Area	% of Site
Native Planting / Biodiversity Strip	Green	1,088 sqm	7.3%
Quiet Contemplation Garden	Passive	884 sqm	5.9%
Commercial & Service Kiosk Cluster	Commercial	748 sqm	5.0%
Multipurpose Sports Lawn	Active	1,292 sqm	8.6%
Perimeter Shade Buffer (N)	Green_Buffer	1,008 sqm	6.7%
Perimeter Shade Buffer (S)	Green_Buffer	1,008 sqm	6.7%
Path Network & Landscape Setbacks (between rooms, to entrances, perimeter jogging loop)	Circulation	3,220 sqm	21.5%

5.3 Circulation

Mode	Strategy
Pedestrians	Primary: 10m-wide Shaded Spine (continuous overhead shade structure) connecting both entrances. Secondary: perimeter jogging/walking loop (dashed, ~4m inset from boundary) plus cross-links into each activity room.
Cyclists	Shared use of the perimeter loop outside peak pedestrian hours; dedicated cycle parking at both entrance plazas (brief requirement, Section F).
Service access	Via the two entrance plazas, routed behind Commercial & Service Kiosk Cluster to avoid crossing main pedestrian flows.
Emergency access	Both entrance plazas sized (12m x 20m) to permit emergency vehicle access; Shaded Spine width (10m) accommodates emergency vehicle passage if required.

5.4 Activity Areas

Active (Children's Play, Outdoor Fitness, Multipurpose Sports Lawn), Passive (Family Picnic, Quiet Contemplation Garden), Social (Community Plaza & Event Lawn), and Green (Native Planting/Biodiversity Strip) zones are distributed on both sides of the Shaded Spine so every activity type has direct shaded access — addressing Phase 2.2's shade-equity problem directly.

5.5 Green Network

Native Planting/Biodiversity Strip (1,088 sqm) plus two Perimeter Shade Buffers (1,008 sqm each, north and south edges) total 3,104 sqm of dedicated green infrastructure (~20.7% of site), rebalancing the existing west-only canopy concentration identified in Phase 1.02/1.11 into a site-wide green network.

5.6 Water Strategy

Drip/subsurface irrigation for all planting zones (per Phase 1.05's near-zero summer rainfall finding); drinking fountains at both entrance plazas and the Community Plaza; no ornamental water features proposed, consistent with Dubai's water-scarcity context and the brief's water-sensitive design requirement.

5.7 Accessibility

100% step-free circulation along the Shaded Spine and all room entries; the perimeter loop and cross-links are graded for wheelchair/stroller use; both entrance plazas provide accessible drop-off adjacency — directly resolving Phase 1.10's undocumented-baseline finding with an explicit universal-design commitment.

5.8 Safety Strategy

Continuous lighting along the Shaded Spine and perimeter loop for night use (brief requires day/night activation); clear sightlines maintained across each room (no fully enclosed blind corners); Community Plaza positioned centrally for natural surveillance from surrounding rooms.

5.9 Wayfinding

Two legible entrances (west Main Plaza, east Secondary Plaza) with the Shaded Spine acting as the primary orientation device — from any point on the spine, every room is a single, short, shaded detour away.

5.10 Smart Infrastructure

- Environmental sensors (temperature/shade monitoring) along the Shaded Spine to validate the Phase 7 shade-performance model against real conditions post-construction.
- Smart irrigation controllers tied to soil-moisture sensors in all green zones.
- Digital wayfinding kiosks at both entrance plazas.

Data Integrity Note

All areas in the schedule above are computed directly from the geometry defined in the script below — not estimated by eye. The exact site boundary (vs. this assumed 150m x 100m rectangle) should be validated once the DWG conversion (logged in 00_MASTER_TRACKER) is complete.

Appendix — Program Proof (Source Code)

The analysis, figures, and tables in this report were generated by the Python script **05_PHASE5_MASTERPLAN_DEVELOPMENT/_scripts/01_generate_masterplan_geometry.py** below. Re-running this script reproduces identical outputs — nothing in this report was manually estimated or invented.

```
"""
Phase 5 - Master Plan Development
Generates an actual geometric zoning + circulation layout for Al Safa 2 Park
implementing the selected "Shaded Spine" concept (Phase 4), sized to the real
15,000 sqm site area confirmed in the Competition Brief.

Site assumed as an elongated rectangle (consistent with the aerial massing seen
in the master-plan graphic, Phase 1.02) since the DWG's exact boundary polygon
is not yet available (logged gap). Proportions: 150m x 100m = 15,000 sqm,
oriented long-axis NW-SE per the site graphic.
"""

import os
import json
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as patches

OUT = os.path.join(os.path.dirname(__file__), "..", "outputs")
os.makedirs(OUT, exist_ok=True)

SITE_W, SITE_H = 150.0, 100.0 # meters; 150 x 100 = 15,000 sqm exactly
SITE_AREA = SITE_W * SITE_H
assert SITE_AREA == 15000, "Site area must match brief's confirmed 15,000 sqm"

# --- Zoning layout (Shaded Spine concept) ---
# The "Spine" runs along the long (east-west, 150m) axis at mid-depth (y=45-55m, 10m wide shaded walkway).
# Rooms are arranged along both sides of the spine.
zones = [
# name, x, y, w, h, color, category
("Main Entrance Plaza", 0, 40, 12, 20, "#C9A24A", "Arrival"),
("Shaded Spine (Central Walkway)", 12, 45, 126, 10, "#3D5A80", "Circulation"),
("Secondary Entrance (E)", 138, 40, 12, 20, "#C9A24A", "Arrival"),

("Children's Play Zone", 14, 58, 32, 34, "#EE6C4D", "Active"),
("Family Picnic & Shaded Seating", 48, 58, 26, 34, "#7FB069", "Passive"),
("Community Plaza & Event Lawn", 76, 58, 36, 34, "#F4A261", "Social"),
("Outdoor Fitness & Wellness", 114, 58, 24, 34, "#4A7C59", "Active"),

("Native Planting / Biodiversity Strip", 14, 8, 32, 34, "#2D6A4F", "Green"),
("Quiet Contemplation Garden", 48, 8, 26, 34, "#8AA29E", "Passive"),
("Commercial & Service Kiosk Cluster", 76, 8, 22, 34, "#B08968", "Commercial"),
("Multipurpose Sports Lawn", 100, 8, 38, 34, "#588157", "Active"),

# Secondary green buffers filling perimeter margins between rooms and site edge
("Perimeter Shade Buffer (N)", 12, 92, 126, 8, "#40916C", "Green_Buffer"),
("Perimeter Shade Buffer (S)", 12, 0, 126, 8, "#40916C", "Green_Buffer"),
("Path Network & Landscape Setbacks", 0, 0, 0, 0, "none", "Path_Remainder"),

("Jogging Track (perimeter loop)", 0, 0, 150, 100, "none", "Circulation_Loop"),
]
```

```
# --- Area accounting (validate against 15,000 sqm total site) ---
area_rows = []
total_room_area = 0
for name, x, y, w, h, color, cat in zones:
if cat in ("Circulation_Loop", "Path_Remainder"):
continue # computed below / overlay only
area = w * h
total_room_area += area
area_rows.append((name, cat, round(area, 1), round(100 * area / SITE_AREA, 1)))

remaining = SITE_AREA - total_room_area # named paths + landscape setbacks between rooms
area_rows.append(("Path Network & Landscape Setbacks (between rooms, to entrances, perimeter jogging loop)",
"Circulation", round(remaining, 1), round(100 * remaining / SITE_AREA, 1)))

print(f"Site area: {SITE_AREA} sqm")
print(f"Total zoned room area: {total_room_area} sqm")
print(f"Remaining (paths/buffers): {remaining} sqm")
for r in area_rows:
print(r)

with open(os.path.join(OUT, "zoning_area_schedule.json"), "w") as f:
json.dump({"site_area_sqm": SITE_AREA, "zones": area_rows}, f, indent=2)
print("Saved: zoning_area_schedule.json")

# --- Draw the master plan diagram ---
fig, ax = plt.subplots(figsize=(15, 10))
ax.set_xlim(-5, SITE_W + 5)
ax.set_ylim(-5, SITE_H + 5)
ax.set_aspect("equal")

# Site boundary
ax.add_patch(patches.Rectangle((0, 0), SITE_W, SITE_H, fill=False, edgecolor="black", linewidth=2.5))

for name, x, y, w, h, color, cat in zones:
if cat == "Path_Remainder":
continue # not drawn as a block; represented by the visible gaps + loop path
if cat == "Circulation_Loop":
# draw perimeter jogging loop (inset from boundary)
inset = 4
loop = patches.Rectangle((inset, inset), SITE_W - 2*inset, SITE_H - 2*inset,
fill=False, edgecolor="#264653", linewidth=2, linestyle="--")
ax.add_patch(loop)
continue
rect = patches.Rectangle((x, y), w, h, facecolor=color, edgecolor="black",
linewidth=1, alpha=0.75)
ax.add_patch(rect)
text_color = "white" if cat not in ("Arrival",) else "black"
ax.text(x + w/2, y + h/2, name, ha="center", va="center", fontsize=7.2,
wrap=True, color=text_color, fontweight="bold")

# North arrow + scale
ax.annotate("N", xy=(SITE_W + 2, SITE_H - 5), fontsize=14, fontweight="bold", ha="center")
ax.annotate("", xy=(SITE_W + 2, SITE_H - 8), xytext=(SITE_W + 2, SITE_H - 15),
arrowprops=dict(arrowstyle="-|>", lw=2))
ax.plot([0, 20], [-3, -3], color="black", linewidth=2)
```

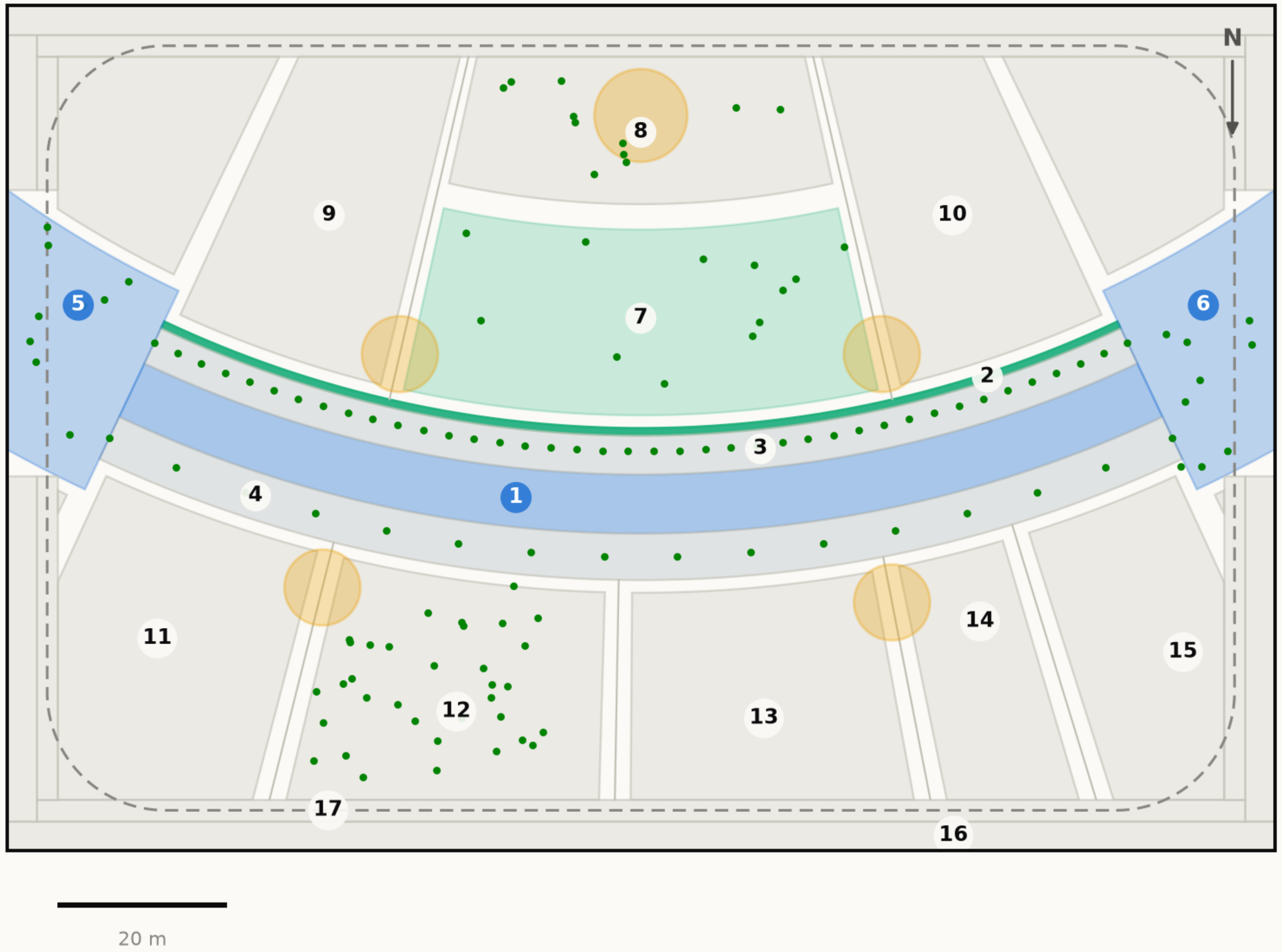
```
ax.text(10, -4.5, "20 m", ha="center", fontsize=9)
ax.set_title("Al Safa 2 Park — Preliminary Master Plan (Concept A: \"Shaded Spine\")\n"
"15,000 sqm | AI-generated draft layout for review", fontsize=13, fontweight="bold")
ax.axis("off")
plt.tight_layout()
fig.savefig(os.path.join(OUT, "masterplan_diagram.png"), dpi=170)
plt.close(fig)
print("Saved: masterplan_diagram.png")
```

Fig10 masterplan

Technical drawing — to scale. Geometry generated by src/plan.py; every area is the measured area of the drawn polygon.

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



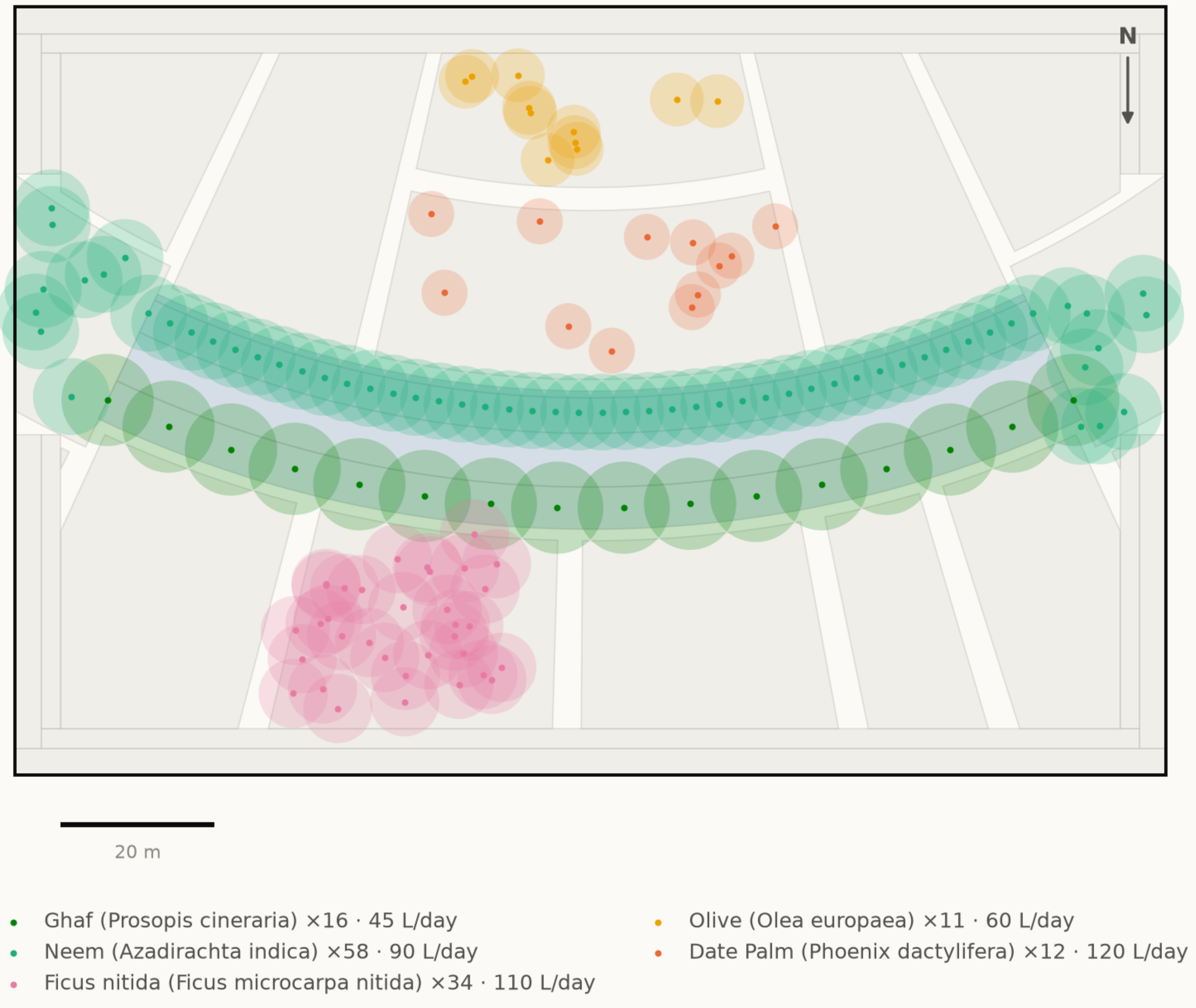
ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
	Al Sikkak — shaded alleys & setbacks	1,414 m²
	Total	**15,000 m²**

Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Planting plan — 131 trees

Drawn at mature canopy radius. The avenue is structural: it carries the shoulder seasons the gridshell cannot



28 of 131 trees are UAE natives (Ghaf and Date Palm). Peak summer irrigation demand 11.8 m³/day. Ghaf takes the southern rank because it is the most drought-tolerant species in the schedule and that rank is the more exposed of the two.

Source: data/raw/species_water_carbon_rates.csv; layout from src/solar.py · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge